

Model Lab

Every prior, same slate — 2026-09-29

SERVED PRIOR

Served (full season)

PRIORS COMPARED

11

HELD-OUT PA

38,539

Held-Out Scores

Fitted on plate appearances before 2026-08-20, scored on the 38,539 that came after (actual HR rate 2.99%). No outcome in the test window was visible during fitting.

Metrics are per plate appearance; the probabilities lower down are per game.

Read AUC first — the slate ranks hitters and takes the top six, so ordering is what matters. Brier and log loss are dominated by the ~3% base rate and barely separate the priors.

Prior	AUC	Top-decile lift	Brier	Log loss	Cal. RMSE
Flat league League HR rate as the prior for every hitter.	0.5837	1.57×	0.02899	0.13366	0.00505
KNN comps Prior is the pooled rate of the nearest comparables.	0.5937	1.49×	0.02897	0.13302	0.00432
SVR Support-vector regression on the same feature matrix.	0.5876	1.47×	0.02906	0.13432	0.00793
Random forest Forest regression; a learned similarity metric.	0.5881	1.50×	0.02903	0.13362	0.00660
Core (KNN+SVR) Blend of KNN and SVR, no forest.	0.5937	1.49×	0.02897	0.13302	0.00432
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.5937	1.49×	0.02897	0.13302	0.00432
Linear regression Ordinary least squares on the same features.	0.5903	1.50×	0.02901	0.13358	0.00541
Logistic regression Binomial logistic on the per-PA home run outcome.	0.5920	1.56×	0.02906	0.13371	0.00750
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.5909	1.53×	0.02901	0.13341	0.00663
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.5889	1.53×	0.02906	0.13391	0.00734

Blend weights (forest variant): KNN 1.00, SVR 0.00, RF 0.00

Shrinkage K per prior: league 257, knn 257, svr 257, rf 257, core 257, ensemble 257, linear 257, logistic 257, nn 257, xgb 257 — estimated by method of moments on out-of-fold residuals, not hand-tuned.

Thin-Sample Subset

The same held-out window, restricted to hitters with fewer than 100 plate appearances before the cutoff (4,464 PA, actual HR rate 2.46%). For a regular with 600 PA the observed record swamps any prior, so the priors mostly separate here.

Prior	AUC	Top-decile lift	Brier	Log loss	Cal. RMSE
Flat league League HR rate as the prior for every hitter.	0.5346	1.46×	0.02406	0.11600	0.00997
KNN comps Prior is the pooled rate of the nearest comparables.	0.5438	1.09×	0.02419	0.11773	0.01438
SVR Support-vector regression on the same feature matrix.	0.5529	1.09×	0.02473	0.12552	0.02726
Random forest Forest regression; a learned similarity metric.	0.5582	1.18×	0.02453	0.12120	0.02355
Core (KNN+SVR) Blend of KNN and SVR, no forest.	0.5438	1.09×	0.02419	0.11773	0.01438
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.5438	1.09×	0.02419	0.11773	0.01438
Linear regression Ordinary least squares on the same features.	0.5560	1.18×	0.02445	0.12185	0.02053
Logistic regression Binomial logistic on the per-PA home run outcome.	0.5572	1.09×	0.02472	0.12166	0.02618
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.5665	1.09×	0.02432	0.11908	0.01892
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.5687	1.09×	0.02476	0.12278	0.02855

Slate Agreement

Served (full season) · 6/6 shared 2/6 hit

Fernando Tatis Jr. ✓
Munetaka Murakami
Pete Crow-Armstrong
Ben Rice ✓
Miguel Vargas
Ronald Acuña Jr.

KNN comps · 6/6 shared 2/6 hit

Fernando Tatis Jr. ✓
Munetaka Murakami
Pete Crow-Armstrong
Ben Rice ✓
Ronald Acuña Jr.
Miguel Vargas

Random forest · 6/6 shared 2/6 hit

Munetaka Murakami
Pete Crow-Armstrong
Fernando Tatis Jr. ✓
Ben Rice ✓
Ronald Acuña Jr.
Miguel Vargas

Forest blend · 6/6 shared 2/6 hit

Fernando Tatis Jr. ✓
Munetaka Murakami
Pete Crow-Armstrong
Ben Rice ✓
Ronald Acuña Jr.
Miguel Vargas

Logistic regression · 6/6 shared 2/6 hit

Flat league · 6/6 shared 2/6 hit

18.9%	Fernando Tatis Jr. ✓	18.6%
18.4%	Pete Crow-Armstrong	17.4%
18.2%	Munetaka Murakami	17.0%
17.5%	Ronald Acuña Jr.	16.1%
16.8%	Ben Rice ✓	16.1%
16.8%	Miguel Vargas	15.6%

SVR · 6/6 shared 2/6 hit

19.2%	Munetaka Murakami	19.1%
18.5%	Pete Crow-Armstrong	18.9%
18.2%	Fernando Tatis Jr. ✓	18.6%
17.4%	Ben Rice ✓	18.2%
16.9%	Ronald Acuña Jr.	17.1%
16.9%	Miguel Vargas	17.0%

Core (KNN+SVR) · 6/6 shared 2/6 hit

18.7%	Fernando Tatis Jr. ✓	19.2%
18.6%	Munetaka Murakami	18.5%
18.6%	Pete Crow-Armstrong	18.2%
17.8%	Ben Rice ✓	17.4%
16.9%	Ronald Acuña Jr.	16.9%
16.9%	Miguel Vargas	16.9%

Linear regression · 6/6 shared 2/6 hit

19.2%	Fernando Tatis Jr. ✓	19.2%
18.5%	Munetaka Murakami	18.3%
18.2%	Pete Crow-Armstrong	18.3%
17.4%	Ben Rice ✓	17.5%
16.9%	Miguel Vargas	17.1%
16.9%	Ronald Acuña Jr.	16.8%

Fernando Tatis Jr. ✓	19.3%	Neural net (torch) · 6/6 shared 2/6 hit	
Pete Crow-Armstrong	18.9%	Fernando Tatis Jr. ✓	19.3%
Munetaka Murakami	18.5%	Pete Crow-Armstrong	18.7%
Ben Rice ✓	17.7%	Munetaka Murakami	18.6%
Miguel Vargas	17.3%	Ben Rice ✓	17.7%
Ronald Acuña Jr.	17.0%	Miguel Vargas	17.2%
		Ronald Acuña Jr.	17.1%
XGBoost · 6/6 shared 2/6 hit			
Munetaka Murakami	19.0%		
Fernando Tatis Jr. ✓	18.8%		
Pete Crow-Armstrong	18.5%		
Ben Rice ✓	17.7%		
Miguel Vargas	17.2%		
Ronald Acuña Jr.	16.8%		

Hitter by Hitter

Hitter	Served (full season)	Flat league	KNN comps	SVR	Random forest	Core (KNN+SVR)	Forest blend	Linear regression	Logistic regression
New York Yankees · vs LHP · 41 HR / 667 PA	17.5%	16.1%	17.4%	18.2%	17.8%	17.4%	17.4%	17.5%	17.7%
5. Miguel Vargas Chicago White Sox · vs RHP · 33 HR / 704 PA	16.8%	15.6%	16.9%	17.0%	16.9%	16.9%	16.9%	17.1%	17.3%
6. Ronald Acuña Jr. slate Atlanta Braves · vs LHP · 20 HR / 469 PA	16.8%	16.1%	16.9%	17.1%	16.9%	16.9%	16.9%	16.8%	17.0%
7. Matt Olson Atlanta Braves · vs LHP · 41 HR / 698 PA	16.0%	14.9%	16.0%	16.8%	16.4%	16.0%	16.0%	16.2%	16.9%
8. Jackson Merrill San Diego Padres · vs LHP · 26 HR / 664 PA	15.9%	14.6%	15.9%	16.1%	16.2%	15.9%	15.9%	16.1%	16.5%
9. Yordan Alvarez slate Houston Astros · vs LHP · 42 HR / 691 PA	15.7%	14.2%	15.7%	16.6%	16.3%	15.7%	15.7%	16.0%	16.8%
10. Manny Machado San Diego Padres · vs LHP · 30 HR / 677 PA	15.4%	14.8%	15.4%	16.0%	15.8%	15.4%	15.4%	15.7%	15.8%
11. Wilyer Abreu slate Boston Red Sox · vs RHP · 24 HR / 681 PA	15.3%	14.4%	15.2%	15.5%	15.5%	15.2%	15.2%	15.3%	15.6%
12. Kyle Schwarber Philadelphia Phillies · vs LHP · 45 HR / 701 PA	15.3%	14.0%	15.3%	16.0%	15.8%	15.3%	15.3%	15.6%	16.5%
13. Kyle Teel Chicago White Sox · vs RHP · 5 HR / 144 PA	14.9%	14.4%	14.5%	14.6%	14.5%	14.5%	14.5%	14.6%	13.7%
14. Jeremy Peña Houston Astros · vs LHP · 22 HR / 483 PA	14.7%	14.5%	14.6%	15.3%	15.0%	14.6%	14.6%	14.9%	14.9%
15. Drake Baldwin Atlanta Braves · vs LHP · 25 HR / 615 PA	14.6%	13.7%	14.5%	14.8%	14.8%	14.5%	14.5%	14.7%	14.6%
16. Willson Contreras Boston Red Sox · vs RHP · 27 HR / 536 PA	14.3%	13.0%	14.4%	14.5%	14.6%	14.4%	14.4%	14.7%	15.4%
17. Alex Bregman Chicago Cubs · vs RHP · 28 HR / 702 PA	14.1%	14.0%	14.1%	14.3%	14.2%	14.1%	14.1%	14.2%	14.2%
18. Ty France San Diego Padres · vs LHP · 21 HR / 500 PA	14.0%	13.5%	14.0%	14.1%	14.1%	14.0%	14.0%	14.0%	14.0%

Hitter	Served (full season)	Flat league	KNN comps	SVR	Random forest	Core (KNN+SVR)	Forest blend	Linear regression	Logistic regression
19. Seiya Suzuki Chicago Cubs · vs RHP · 26 HR / 634 PA	13.9%	13.5%	13.9%	14.2%	14.2%	13.9%	13.9%	13.9%	14.2%
20. Adley Rutschman Boston Red Sox · vs RHP · 13 HR / 410 PA	13.6%	13.4%	13.7%	13.4%	13.0%	13.7%	13.7%	13.5%	13.2%
21. Andrew Benintendi Chicago White Sox · vs RHP · 15 HR / 468 PA	13.4%	12.4%	13.3%	13.1%	13.0%	13.3%	13.3%	13.5%	13.7%
22. Paul Goldschmidt New York Yankees · vs LHP · 18 HR / 380 PA	13.1%	12.6%	13.1%	13.8%	12.9%	13.1%	13.1%	13.5%	13.2%
23. Heliot Ramos New York Yankees · vs LHP · 13 HR / 442 PA	13.1%	12.4%	13.1%	12.6%	12.5%	13.1%	13.1%	13.1%	12.9%
24. Austin Riley Atlanta Braves · vs LHP · 19 HR / 630 PA	12.8%	11.8%	12.7%	12.9%	12.3%	12.7%	12.7%	12.5%	12.7%
25. Spencer Jones New York Yankees · vs LHP · 11 HR / 289 PA	12.6%	11.5%	12.5%	11.6%	11.9%	12.5%	12.5%	11.8%	11.9%
26. Lucas Spence Houston Astros · vs LHP · 4 HR / 63 PA	12.5%	10.8%	12.6%	12.2%	12.9%	12.6%	12.6%	12.8%	12.5%
27. Derek Hill Philadelphia Phillies · vs LHP · 11 HR / 216 PA	12.5%	12.3%	12.5%	13.2%	13.0%	12.5%	12.5%	13.1%	12.5%
28. Bryce Harper Philadelphia Phillies · vs LHP · 29 HR / 693 PA	11.8%	11.1%	11.8%	12.1%	11.8%	11.8%	11.8%	11.8%	11.9%
29. Jarren Duran Boston Red Sox · vs RHP · 20 HR / 608 PA	11.8%	11.2%	11.9%	12.2%	11.7%	11.9%	11.9%	12.0%	12.2%
30. Michael Busch Chicago Cubs · vs RHP · 20 HR / 710 PA	11.7%	11.1%	11.6%	11.1%	11.3%	11.6%	11.6%	11.6%	11.3%
31. Colson Montgomery Chicago White Sox · vs RHP · 31 HR / 638 PA	11.7%	10.4%	11.7%	11.3%	11.6%	11.7%	11.7%	11.8%	12.4%
32. Christian Walker Houston Astros · vs LHP · 29 HR / 635 PA	11.5%	11.5%	11.4%	12.1%	11.7%	11.4%	11.4%	11.7%	11.9%
33. Yainer Diaz	11.4%	11.6%	11.1%	11.9%	11.3%	11.1%	11.1%	11.6%	11.5%

Hitter	Served (full season)	Flat league	KNN comps	SVR	Random forest	Core (KNN+SVR)	Forest blend	Linear regression	Logistic regression
Houston Astros · vs LHP · 14 HR / 361 PA									
34. Austin Hays San Diego Padres · vs LHP · 3 HR / 105 PA	11.4%	11.6%	11.2%	10.7%	11.2%	11.2%	11.2%	11.6%	10.8%
35. George Lombard Jr. New York Yankees · vs LHP · 7 HR / 166 PA	11.3%	11.5%	11.2%	11.6%	11.4%	11.2%	11.2%	11.6%	10.8%
36. Isaac Paredes Houston Astros · vs LHP · 25 HR / 659 PA	11.1%	11.4%	10.9%	11.0%	11.2%	10.9%	10.9%	11.0%	11.0%
37. Cody Bellinger New York Yankees · vs LHP · 17 HR / 562 PA	11.0%	11.4%	10.9%	10.8%	10.9%	10.9%	10.9%	11.0%	11.0%
38. Tristan Peters Chicago White Sox · vs RHP · 12 HR / 491 PA	10.9%	11.3%	10.8%	10.7%	11.1%	10.8%	10.8%	11.0%	11.1%
39. Cam Smith Houston Astros · vs LHP · 21 HR / 568 PA	10.9%	10.4%	11.0%	11.3%	11.1%	11.0%	11.0%	11.0%	11.0%
40. Ian Happ Chicago Cubs · vs RHP · 24 HR / 636 PA	10.8%	9.9%	10.9%	10.6%	10.8%	10.9%	10.9%	10.8%	10.8%

Per-game probabilities: each prior's per-PA rate run through this hitter's own plate appearances, park, weather, bullpen and times-through-the-order terms. Only the prior changes across the columns — every other term is held fixed, so a left-to-right spread is the prior disagreeing about the hitter, not the context.